MILITARY
SPECIFICATION

MIL-R-11764A(SigC)
29 December 1960
Superseding
MIL-R-11764(SigC)
18 February 1952

RESISTOR, DECADE ZM-16( )/U

1. SCOPE

1.1 This specification covers one type of variable resistance device, for varying resistance in steps between 0.1 and 111, 111, 111 ohms; designated Resistor, Decade ZM-16( )/U.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

UU-T-111	Tape, Paper, Gummed (Sealing and Securing)
QQ-S-571	Solder: Lead Alloy, Tin Lead Alloy, and Tin Alloy, Flux Cored Ribbon and Wire, and Solid Form-Q
PPP-B-566	Boxes, Folding, Paperboard
PPP-B-585	Boxes, Wood, Wirebound
PPP-B-591	Boxes, Fiberboard, Wood-Cleated
PPP-B-601	Boxes, Wood, Cleated Plywood
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner
PPP-B-636	Boxes, Fiber
PPP-P-291	Paperboard, Wrapping, Cushioning

MILITARY

MIL-I-10	Insulating Materials, Electrical, Ceramic, Class L
MIL-M-14	Molding Plastics and Molded Plastic Parts, Thermosetting
MIL-P-116	Preservation, Methods of

FSC 6625

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MIL-B-117	Bags, Interior Packaging
MIL-B-131	Barrier-Material; Water Vaporproof, Flexible
MIL-V-173	Varnish, Moisture and Fungus-Resistant for the Treatment of Communications, Electronic, and Associated Electrical Equipment
MIL-I-631	Insulation, Electrical, Synthetic-Resin Composition, Nonrigid
JAN-A-669	Anti-Seize Compound, White Lead Base, General Purpose (for Threaded Fittings)
MIL-I-3190	Insulation Sleeving, Electrical, Flexible, Treated
MIL-I-7444	Insulation Sleeving, Electrical, Flexible
MIL-T-7928	Terminals, Lug, Crimp Style, Copper, Insulated
MIL-B-10377	Box, Wood, Cleated, Veneer, Paper-Overlaid
MIL-R-10509	Resistors, Fixed, Film (High Stability)
MIL-L-10547	Liners, Case, Waterproof
MIL-F-14072	Finishes for Ground Signal Equipment
MIL-F-14256	Flux, Soldering, Liquid (Rosin Base)
MIL-E-15090	Enamel, Equipment, Light-Gray (Formula No. 11)
MIL-T-15659	Terminals; Lug Solder Type, Copper
MIL-I-18057	Insulation Sleeving, Electrical, Flexible, Glass Fiber, Silicone Rubber Treated

STANDARDS

MILITARY

MIL-STD-105	Sampling for Expensive Testing by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-169	Extreme-Temperature Cycle
MIL-STD-170	Moisture Resistance Test Cycle for Ground Signal Equipment
MIL-STD-252	Wired Equipment, Classification of Visual and Mechanical Defects
MS-35436	Terminal, Lug, Solder Type, Copper Stamping, Insulation Grip, One Hole
MS-35440	Terminal, Lug, Solder Type, Copper Stamping, Insulation Grip, Solder

DRAWINGS

SIGNAL CORPS

SC-DL-188507	Resistor Decade ZM-16( )/U

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(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions may be obtained from or as directed by the contracting officer. Both the title and identifying number or symbol should be stipulated when requesting copies).

2.2 Other publications.- The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

Consolidated Classification Committee
Uniform Freight Classification No. 4
Consolidated Freight Classification, Rule 21

(Application for copies should be addressed to the Consolidated Classification Committee, 202 Union Station, Chicago 6, Ill.).

American Trucking Association, Inc.
National Motor Freight Classification No. A-3 and No. 14

(Application for copies should be addressed to the National Classification Board, 1424 Sixteenth Street, NW. Washington 6, D. C.).

3. REQUIREMENTS

3.1 Description.- The decade resistor is a single portable unit that provides fixed, accurate, conveniently selectable resistance values by means of nine decade resistance units. Each unit consists of ten resistors, connected in series, and a rotary decade switch. Resistance ranges of the switches are 0 to 1.0, 0 to 10, 0 to 100, 0 to 1,000, 0 to 10,000, 0 to 100,000, 0 to 1,000,000, 0 to 10,000,000 and 0 to 100,000,000 ohms.

3.2 Preproduction samples.- The contractor shall furnish two preproduction samples of complete items and one of each decade unit type for approval, if required by the invitation for bids and contract. (See 4.3)

3.3 Construction.- The equipment shall be constructed in accordance with Drawing and Data List SC-DL-188507.

3.3.1 Adjustment and repair.- The equipment shall be so constructed that parts, terminals, wiring, etc, are accessible for circuit checking, adjustment, maintenance, repair, and replacement with minimum disturbance to other parts and wiring and with use of the minimum number and variety of special tools.

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3.3.2 Clearance and creepage.- The minimum electrical clearance shall be 1/8 inch and the minimum electrical creepage shall be 3/8 inch. (See 4.18)

3.3.3 Interchangeability.- Like units, assemblies, subassemblies and replaceable parts shall be physically and functionally interchangeable, without modification of such items or of the equipment. (See 4.7) Individual items shall not be hand-picked for fit or performance, however, matched pairs or sets, when permitted, may be interchangeable as such.

3.3.4 Safety measures.-

3.3.4.1 Mechanical.- Edges and corners of enclosures shall be rounded. (See 4.18)

3.3.4.2 Electrical.- With the equipment assembled and set up for operation, personnel shall be protected from contact with potentials in excess of 30 volts. (See 4.18)

3.4 Adhesives.- Where adhesives are necessary to attach plastic, rubber, or synthetic rubber to plastic or metal, the adhesive shall not flow, lose its adhering strength or any other favorable characteristics when exposed to any of the environmental conditions or tests covered herein. Excess adhesive shall be removed. (See 4.18)

3.5 Lubricants.- Equipment shall be completely operable and function satisfactorily throughout its normal life without addition of lubricant. (See 4.18)

3.6 Metals, dissimilar.- Where dissimilar metals are in contact, they shall be protected as required by Specification MIL-F-14072. (See 4.18)

3.7 Plastics.- Where not machined, plastic material and parts, such as case and panel, shall have the original smooth or polished surfaces. Surfaces that have been sawed, cut, punched, or otherwise machined shall be as smooth as practicable in accordance with good manufacturing practice for the intended application. Plastic parts shall be in accordance with applicable drawings covered in Drawing and Data List SC-DL-188507. (See 4.18)

3.8 Cleaning.-

3.8.1 Parts.- After fabrication, parts shall be cleaned in accordance with good commercial (See 4.17)

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practice, or as specified in an applicable document. Cleaning processes shall have no deleterious effect. Corrosive material shall be removed completely before the parts are mounted on chassis, panels, etc. (See 4.18)

3.8.2 Units.- After assembly, units shall be cleaned thoroughly and shall be free from particles of solder, flux, and other foreign material. In addition, when necessary, such cleaning shall also be performed before final assembly of the units. (See 4.18)

3.9 Finish, protective.- Equipment shall be given protective finish in accordance with Specification MIL-F-14072. This includes finish of hardware, such as handles, hinges, screws, etc, and necessary touch up after mounting. The final paint film on type I surfaces shall be film E, semigloss, light-gray enamel, conforming to Specification MIL-F-14072.

3.10 Marking.-

3.10.1 General.- Marking shall conform to Specification MIL-M-13231. Front panel marking shall be group I as described in the referenced specification. (See 4.18)

3.10.2 Visibility.- Wherever practicable, parts shall be so mounted that their identification markings will be readily visible with minimum disassembly of the equipment. (See 4.18)

3.10.3 Serial numbers.- All ZM-16( )/U decade resistors shall be given serial numbers. (See 4.1)

3.11 Riveting.- Riveting shall not be used for mounting parts such as resistors, and switches. Thickness of countersunk metal which accepts the heads of flush rivets shall be not less than the height of the rivet heads. The distance from the centers of rivet holes to the edges of the material in which the holes are formed shall be at least 1-1/2 times the rivet diameter. After riveting, the joints shall be tight, the joined parts shall be undamaged, and the rivet heads shall be properly seated and tight against their bearing surfaces. (See 4.18)

3.12 Securing of parts.-

3.12.1 General.- Lugs, flanges, inserts, bolts, and other mounting arrangements shall be such as to retain items securely when subjected to specified service conditions. Parts shall be so mounted that loosening, wear of mounting arrangements, or permanent seperation of parts will not occur when the items in which they areused are subjected to specified bounce and shock tests. Any permanent deformation resulting from the tests shall not cause degradation of performance below full specification requirements. When practicable, friction between mating surfaces shall not be employed as the sole means of preventing fixed parts from rotating. Parts shall be secured in such manner that failure of a single rivet or screw will not free the part completely. (See 4.18)

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3.13 Soldering.-

3.13.1 Solder.- Solder shall conform to Specification QQ-S-571. For soldering electrical connections, composition SN60, type AR or S, shall be used for general purposes, and composition SN62 or SN63, type AR or S, for special purposes. When type S solder is used for soldering electrical connections, the flux shall conform to Specification MIL-F-14256.

3.13.2 Acid or acid salts.- No acid or acid salts shall be used in preparation for or during soldering, however, exception is permitted for preliminary tinning of electrical connections and for tinning or soldering of mechanical joints not used to complete electrical circuits, but in no case shall acid or acid salts be used where they can come in contact with insulation material. Where acid or acid salts are used, as permitted above, they shall be completely neutralized and removed immediately after use.

3.13.3 Process.- There shall be no sharp points or rough surfaces resulting from insufficient heating. The solder shall feather out to a thin edge, indicating proper flowing and wetting actions, and shall not be crystallized, overheated, or under heated. The minimum necessary amount of flux and solder shall be used for electrical connections. Any means employed to remove an unavoidable excess of flux shall not incur the risk of loose particles of flux, brush bristles, or other foreign material remaining in the equipment; flux being spread over a larger area; or damage to the equipment. Insulation material that has been subjected to heating during the soldering operation shall be undamaged and parts fastened thereto shall not have become loosened. (See 4.18)

3.14 Tropicalization of material.- Material shall be treated as follows

3.14.1 Treating materials.- Treating materials containing a mercury-bearing fungicide shall not be used. The contractor shall determine that the treating material is compatible with the material or surface to be treated. Selection of treating materials shall be such that any increase in flammability of treated material will be held to the practical minimum.

3.14.2 Toxicity.- Treating materials shall cause no skin irritation or other injury to personnel handling the treated material during fabrication, transportation, operation, or maintenance of the equipment, or during use of the finished items when used for the purpose intended.

3.14.3 Flexibility.- Treatment shall not affect the flexibility of treated materials, to the extent that the equipment may fail to meet specified requirements when subjected to specified service conditions.

3.14.4 Statement of treatment.- The contractor shall submit for approval, to the contracting officer, a statement describing in detail the materials to be treated and the treating materials and processes that he proposes to use.

3.15 Wiring.- Wiring shall be neat and sturdy. (See 4.18)

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3.15.1 Slack.- Wires shall be as short as practical except that sufficient slack shall be provided for the following purposes.

(a) To enable parts to be removed and replaced during servicing without disconnecting other parts.

(b) To ensure freedom of motion of lugs or terminals normally intended to have some degree of movement.

(c) To facilitate field repair of broken or cut wires.

(d) To prevent chafing or breaking of individual wires due to repeated flexing of hinged parts.

3.15.2 Protection.- Wires and cables shall be so placed and protected as to avoid contact, under specified service conditions, with rough or irregular surfaces or sharp edges. Wires shall not be bent sharply where they enter insulation material.

3.15.3 Support.- Wire shall be properly supported and secured, to prevent undue stress on the conductors and terminals and undue change in position of the wire or cable (i) during and after subjection of the equipment to specified service conditions or (ii) after service or repair of the equipment in a normal manner.

3.15.4 Splicing.- Wires in continuous run between two terminals shall not be spliced during the wiring operation.

3.15.5 Insulating sleeving.- Insulating sleeving shall not be used unless specifically approved by the contracting officer. Approval will not be granted when other means of insulation are practicable. If insulating sleeving is used, it shall conform to Specification MIL-I-631, MIL-I-3190, MIL-I-7444, or MIL-I-18057 as applicable.

3.15.6 Connections.- (See 4.18)

3.15.6.1 General.- Before being soldered to terminal lugs or fixed terminals, wires shall be mechanically secured so that the connections are not dependent for strength on solder alone. Where practicable, wires soldered to fixed terminals shall be looped at least one-half turn but not more than three-quarters turn around the terminal before soldering. Bared ends of wire leads to be terminated in solder-type terminal lugs shall be tinned, silver plated, or lead-alloy coated. Electrical connections shall not be made by clamping between a metallic and non-metallic material. Fraying of textile ends of wires shall be prevented mechanically or by

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application of varnish conforming to Specification MIL-V-173. No varnish, lacquer, inspection paint, or other coating shall be applied to completed electrical connections.

3.15.6.2 Terminal lugs.- Terminal lugs shall be of the type that grips the wire insulation. Solderless terminal lugs shall conform to Specification MIL-T-7928. Solder-type terminal lugs shall conform to Specification MIL-T-15659 and shall be of the types shown on Standards MS35436 and MS35440.

3.16 Anti-seize compound.- Anti-seize compound conforming to Specification JAN-A-669 shall be used to prevent seizing of threads where the threaded surface is aluminum or magnesium.

3.17 Controls.- Controls shall conform to the following requirements:

3.17.1 Effect of vibration, bounce and shock.- Controls shall maintain their setting when the items in which they are used are subjected to specified operating service conditions.

3.17.2 Mechanical operation.- Play and backlash shall not cause poor contact or inaccurate setting. Controls shall operate freely and smoothly without lubrication. (See 4.18)

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3.17.3 Nonturn device.- Controls shall be equipped with a positive device such as a locknut, to prevent their turning in respect to the panel during or upon completion of vibration, bounce and shock tests.

3.18 Decade resistance units.- Each decade resistance unit shall consist of an assembly which includes a switch and ten equal resistance elements. The zero resistance of each decade unit shall be less than 0.005 ohms. The zero resistance of the entire assembly shall be less than 0.05 ohms. The terminal shunt capacitance of the assembly with 100 megohms set on the box shall be less than 10 uuf. (See 4.8 and 4.8.2.1)

3.18.1 Accuracy.- The accuracy of the resistance measured at the terminals of the decade unit with dc and minus the zero resistance, shall be as specified in Table II. (See 4.8.1, 4.8.1.1, and 4.8.1.2)

3.18.2 Maximum power dissipation.- The resistance at each step shall be maintained within specified limits while dissipating the continuous power and maximum current specified in table II. No charring or visual damage shall occur to the decade unit due to such operation. (See 4.8.3)

Table II.- Resistance test requirements

Decade Range (ohms)	Accuracy (±%)	Max. Power (watts)	Max. Current (amps)
0-1	2	2	4.50
0-10	1	0.5	0.71
0-100	1	0.5	0.22
0-1000	1	0.5	0.071
0-10,000	1	0.5	0.022
0-100,000	1	0.5	.0071
0-1 Meg	1	0.5	.0022
0-10 Meg	1	0.5	.0007
0-100 Meg	1	.025	50 Micro

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3.19 Switches.- The switches shall have eleven contacts and be of the shorting type. They shall have stops so that rotation in the counterclockwise direction beyond the zero resistance position is prevented and rotation in the clockwise direction beyond the maximum resistance position is prevented. The switches shall be able to withstand 1000 volts d.c. between adjacent stator terminals without arcing. (See 4.8.4)

3.20 Service conditions.-

3.20.1 Atmospheric.- The decade resistor shall be so constructed that no fixed part or assembly shall become loose, no moving or movable part or assembly shall become undesirably free or sluggish in operation, and no degradation shall be caused in the performance below that specified in the individual specification for the particular test set when subjected to the following conditions.

3.20.1.1 Test set operating.- The test set shall operate under any possible combination of the following ambient conditions.

3.20.1.1.1 Temperature.- Continuous operation over the range of 0°F to +125°F. (See 4.11)

3.20.1.1.2 Altitude.- Operation at barometric pressures ranging down to 20.6 inches of mercury (approximating an altitude of 10,000 feet). (See 4.9)

3.20.1.2 Test set nonoperating.- The following conditions and any possible combination thereof are those to which the decade resistor may be subjected while inoperative. The decade resistor shall sustain no degradation in performance below that specified in this specification for the particular decade resistor after returning to the operating range service conditions.

3.20.1.2.1 Temperature.- Ranging from -80 to 160°F for extended periods. (See 4.11)

3.20.1.2.2 Humidity.- Ranging up to 98 percent relative for extended periods. (See 4.12)

3.20.1.2.3 Altitude.- Barometric pressures ranging down to 3.4 inches of mercury approximating an altitude of 50,000 feet. (See 4.9)

3.20.1.2.4 Fungus.- Subjected for extended periods to fungus and fungus supporting conditions. (See 4.19)

3.20.2 Mechanical.- The decade resistor shall be so constructed that no fixed part shall become loose, no movable part or control shall be shifted in setting, position, or adjustment, and no permanent degradation shall be caused in performance below that specified in the individual specification for the particular test set when subjected to the following conditions.

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3.20.2.1 Operating.-

3.20.2.1.1 Vibration.- The test set shall not suffer damage or fail to give the performance required by the individual specification for the particular test set, when subjected to vibration in accordance with the requirements of 4.17.1 and its subparagraphs.

3.20.2.2 Nonoperating.-

3.20.2.2.1 Vibration.- No impaired performance or damage to the decade resistor shall result from vibrations as specified in 4.17.2 and its subparagraphs.

3.20.2.2.2 Bounce.- No impaired performance or damage to the decade resistor shall result from bouncing specified in 4.16.

3.20.2.2.3 Bench handling.- No impaired performance or damage to the decade resistor shall result from handling specified in 4.14.

3.20.2.2.4 Drop.- No impaired performance or damage to the decade resistor shall result from drops specified in 4.15.

3.21 Electrical.- The resistance of the resistive elements of all the decade units shall remain within the limits specified in paragraph 3.18, and its' subparagraphs, under the conditions specified in the following table:

Table III

Conditions	References
Atmospheric	Paragraph 3.20.1 and its' subparagraphs
Mechanical	Paragraph 3.20.2 and its' subparagraphs
A.C. input (up to 60 cps)	Paragraph 3.18 and its' subparagraphs
DC input	Paragraph 3.18 and its' subparagraphs
*Maximum current	Dwg SM-D-188508

3.22 Life.- The test set shall be designed to have a reliable operating life of at least 200 hours without requiring servicing. Parts requiring servicing at the end of this interval shall be designated by the contractor. (See 4.20)

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3.23 Degree of enclosure.-

3.23.1 Cover closed.- With cover closed and latched, the case shall be immersion proof when immersed in 3 feet of water. Gaskets, or any other device, utilized to ensure against leakage of water, shall not require special adjustment prior or following closure to ensure fulfillment of this requirement. (See 4.10)

3.23.2 Cover open.- With cover open, the panel shall be rain proof. Any device utilized to ensure against entrance of water, shall not require special adjustment to ensure this feature. (See 4.13)

3.24 Preconditioning.- The equipment shall be capable of meeting the requirements herein, without subsequent processing, after subjection to the bounce preconditioning of 4.6. (See 4.5)

3.25 Technical literature.- Technical literature, shall be furnished if specified in the contract.

3.26 Workmanship.- The equipment shall be manufactured and assembled in accordance with the applicable portions of the following paragraphs herein:

3.7 General requirements for plastic material and plastic parts
3.8 Cleaning
3.11 Riveting
3.12.1 General requirements for securing of parts
3.13.2 Flux and cleaning agents for soldering
3.13.3 Process for soldering
3.15 Wiring
3.15.1 Slack in wires
3.15.2 Protection for wires
3.15.4 Splicing of wires
3.15.6.1 Connections, general
3.17.2 Mechanical operation of controls

4. QUALITY ASSURANCE PROVISIONS

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4.1 Contractor's responsibility.- The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspection.- Inspection shall be classified as follows

(a) Preproduction inspection (does not include preparation for delivery). (See 4.3)

(b) Acceptance inspection.

(1) Acceptance inspection of equipment before preparation for delivery. (See 4.4 and 4.5)

(2) Acceptance inspection of preparation for delivery. (See 4.23)

4.3 Preproduction inspection.- This inspection will be performed by the Government unless otherwise specified in the contract. It shall consist of the preproduction inspection specified in table III and the inspection specified in the subsidiary documents covering the items listed in 4.4.

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Table III Preproduction inspection

Inspection	Rqt Par	Insp Par
Visual and mechanical inspection	3.3.1, 3.3.2, 3.3.3, 3.3.4, 3.3.5.1, 3.3.5.2, 3.4, 3.5, 3.6, 3.7, 3.8, 3.9, 3.10, 3.11, 3.12.1, 3.12.2, 3.12.2.1, 3.12.2.2, 3.12.2.3, 3.12.2.4, 3.12.3, 3.13.1, 3.13.2, 3.13.3, 3.14.3, 3.15.1, 3.15.2, 3.15.3, 3.15.4, 3.15.5, 3.15.6.1, 3.15.6.2, 3.15.6.3, 3.17.1, 3.17.2, 3.17.3, 3.19, 3.21.1 and 3.25	4.7 and 4.18
Bounce preconditioning	3.24	4.6
Resistance	3.18, 3.18.1, Table II and 3.21	4.8.1.1, 4.8.1.2
Terminal shunt capacitance	3.18	4.8.2 and 4.8.2.1
Maximum power dissipation	3.18.2	4.8.3
Adjacent stator terminals insulation	3.19	4.8 and 4.8.4.1
High temperature (non operating)	3.20.1, 3.20.1.2 and 3.20.1.2.1	4.11
High temperature (operating)	3.20.1, 3.20.1.1 and 3.20.1.1.1	4.11
Low temperature (non operating)	3.20.1, 3.20.1.2 and 3.20.1.2.1	4.11
Low temperature (operating)	3.20.1, 3.20.1.1 and 3.20.1.1.1	4.11
Altitude (non operating)	3.20.1, 3.20.1.2 and 3.20.1.2.3	4.9
Altitude (operating)	3.20.1, 3.20.1.1, and 3.20.1.1.2	4.9
Fungus	3.20.1, 3.20.1.2, and 3.20.1.2.4	4.19
Humidity	3.20.1, 3.20.1.2, and 3.20.1.2.2	4.12
Vibration (non operating)	3.20.2 and 3.20.2.2.1	4.17.2
Vibration (operating)	3.20.2 and 3.20.2.1.1	4.17.1
Bounce (non operating)	3.20.2 and 3.20.2.2.2	4.16
Bench handling (non operating)	3.20.2 and 3.20.2.2.3	4.14
Drop (non operating)	3.20.2 and 3.20.2.2.4	4.15
Life	3.22	4.20
Immersion	3.23.1	4.10
Rain	3.23.2	4.13

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4.3.1 Test sequence.- Tests to be performed, if feasible, in the listed sequence. Exceptions may be taken as follows.

a. Sequence of power dissipation and stator terminals insulation tests may be reversed.

b. Sequence of fungus and humidity tests may be reversed.

c. The preproduction inspection will normally be performed in this order (1) vibration, (2) bounce, (3) shock, benchhandling, (4) shock, or drop, and (5) immersion, other preproduction inspection may precede, follow, or be interspersed between the foregoing.

d. Life test may follow rain or immersion tests.

4.4 Acceptance inspection covered by subsidiary documents.- The following shall be inspected under the applicable subsidiary documents as part of the acceptance inspection before preparation for delivery:

Item	Where Required	Documents
Finishes	Dwg SC-DL-188507	MIL-F-14072
Marking	Dwg SC-DL-188507	MIL-M-13231
Resistors, fixed	Dwg SC-DL-188507	MIL-R-10509
Switches, rotary	Dwg SC-DL-188507	MIL-S-3786

4.5 Acceptance inspection of equipment before preparation for delivery.- The contractor, to demonstrate compliance with specified requirements, shall perform the inspection specified in 4.4 and 4.5.1 through 4.5.4. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition the Government—at its discretion—may perform all or any part of the specified inspection, to verify the contractor's compliance with specified requirements. (See 6.6) Test equipment for Government verification inspection shall be made available by the contractor. Each unit which will be subjected to group A, group B, or group C inspection shall be preconditioned after final assembly.

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4.5.1 Group A inspection.- This inspection, including sampling, shall conform to table V and the ordinary inspection procedures of Standard MIL-STD-105. Group A inspection may be performed in the order shown in table V.

Table V Group A Inspection

Inspection	Rqt Par	Inspection Par	AQL (DPHU) Major	AQL (DPHU) Minor
Visual and Mechanical	3.24, 3.26	4.18, 4.6	1.0	4.0
Electrical				
Resistance tests				
D.C. Zero (decade units)	3.18	4.8.1)		*
D.C. Zero (assembly)	3.18	4.8.1)		*
Accuracy	3.18.1	4.8.1)		*
Terminal shunt capacitance	3.18	4.8.2)	1.0 for the combined group	*
Stator terminals insulation	3.19	4.8.4)		*

DPHU = Defects per hundred units.
* No minor electrical defects.

4.5.2 Group B inspection.- This inspection, including sampling, shall conform to table VI and to the special procedures for small-sample inspection of Standard MIL-STD-105. The inspection level, for normal and tightened inspection, shall be L-7 for normal and tightened and L-5 for reduced inspection. The reduced inspection procedure shall be R-1. Group B inspection shall normally be performed on inspection lots that have passed group A inspection and on samples selected from units that have been subjected to and met the group A inspection.

4.5.2.1 Order of inspection within group B.- Group B inspection shall be performed in the order shown in table VI.

4.5.2.2 Procedure in case of failure.- When an inspection lot fails, it shall be rejected and the contractor shall immediately investigate the cause of failure and take corrective action to assure that subsequent lots do not contain the same defect or defects.

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Table VI Group B.- Acceptance inspection

Inspection	Rqt Par	Insp Par	AQL (%)
Interchangeability	3.3.4	4.7	6.5
Benchhandling	3.20.2 and 3.20.2.3	4.14)	6.5 (combined)
Power dissipation	3.18.2	4.8.3)	
Immersion	3.23.1	4.10	6.5

4.5.3 Group C inspection.- This inspection shall be as listed in tab VII, and shall normally be performed on sample units that have been subjected to and ; et group A and group B inspection.

Table VII Group C.- Acceptance inspection

Inspection	Rqt Par	Insp Par
Moisture resistance	3.20.1.2.2	4.12

4.5.3.1 Sampling for group C tests.- One decade resistor for every 100 units or fraction thereof produced, or one every month, whichever is sooner, shall be selected for group C tests specified in table VII. The first unit shall be selected from the first units produced at start of the contract.

4.5.3.2 Noncompliance.- If a sample unit fails group C inspection, the contractor shall immediately investigate the cause of failure and shall report to the Government inspector the results thereof and details of the corrective action taken on the process and all units of product which were manufactured with the same conditions, materials, processes, etc. If the Government inspector does not consider that the corrective action will enable the product to meet specified requirements, or if the contractor cannot determine the cause of failure, the matter shall be referred to the contracting officer. (See 6.4)

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4.5.4 Reinspection of conforming group B and group C sample units.- Unless otherwise specified, sample units which have been subjected to and passed group B or group C inspection, or both, may be accepted on contract, provided that they are resubjected to and pass group A inspection after repair of all visible damage.

4.6 Bounce preconditioning.- The unit shall be placed in its normal operation position on the table of the Package Tester as made by the L.A.B. Corporation, Skaneateles, New York, or equal. The package tester, shafts in phase, shall have a speed such that it is just possible to insert a 1/32-inch-thick strip of material under one corner oredge of the unit to a distance of 3 inches as the unit bounces. The unit shall be subjected to this preconditioning for 1 minute. After bounce preconditioning, the unit shall not be repaired, alined, cleaned, or otherwise changed prior to subjection to acceptance inspection.

4.7 Inspection for interchangeability.- The dimensions listed below shall be gaged or measured to determine conformance to the physical interchangeability requirement of 3.3.4. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

(a) External and internal dimensions of case and cover.

(b) Location of hinges and fasteners.

(c) Size and form of special threads.

4.8 Electrical tests.-

4.8.1 Resistance tests.- The resistance of each resistance value available from the decade shall be measured at the output terminals with an acceptable resistance measuring instrument. The zero resistance of each decade unit, with the decade resistance unit switch set at zero, shall be measured to include all the resistance between a switch terminal on one switch to the opposite switch terminal of the next switch in the decade circuit. The zero resistance of the overall assembly shall include all ten of these resistances and the leads and contact resistance from the outlet connectors to the X100,000 ohm and X10 megohm switches.

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4.8.1.1 Resistance measuring instruments.- The full scale or overall accuracy of the d.c. resistance measuring instruments used to check resistance accuracy shall be accurate within the following percentages:

Range	Accuracy (Percent)
Up to and including 1 ohm	±0.2
Above 1 ohm	±0.1

Where scales are a part of the resistance measuring device, only the upper half shall be used. When measuring the resistance, care shall be taken to avoid overloading the resistors. The wattage ratings herein and the "short time overload" tests in Specification MIL-R-10509 shall be utilized as guidelines in taking necessary measurements.

4.8.1.2 Preconditioning resistance test samples.- After the initial visual and mechanical inspection, any sample that is to be subjected to the resistance test, is to be preconditioned as follows:

a. 130 ±10°F for 10 hours
b. Remove lint or deposit upon completion with forced air

4.8.2 Capacitance test.- The shunt capacitance of the assembly shall be measured at the terminals with a recently calibrated, acceptable capacitance measuring instrument.

4.8.2.1 Preconditioning capacitance test samples.- If this test is conducted on the resistance test samples within 72 hours following the resistance test, they will not have to be preconditioned. However, if the time interval exceeds this period, the preconditioning covered in paragraph 4.8.1.2 will be repeated prior to capacitance.

4.8.3 Power dissipation test.- The voltage shall be increased at the top step of each decade switch so that the maximum power requirement is reached on each decade switch with ten unit resistors in series. This voltage shall be applied for at least 5 minutes on each unit individually. Resistors and adjacent components shall be inspected during and following loaded conditions to ascertain if resistors have charred or any other visual damage has occurred to the resistors or other components. This test may be combined with the resistance test to save time. The resistance shall be measured at the same 10 dial positions upon completion of the power dissipation tests to ascertain if resistance is still within tolerance specified.

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4.8.4 Adjacent stator terminal insulation test.- Each pair of adjacent stator terminals shall be subjected to 1000 +100 -0 volts, d-c, for a period of 10 +2 -0 seconds, after electrical isolation from other circuit components. The equipment shall meet the following requirements:

a. No arcing during application of voltage.
b. No visible deterioration of insulation or any other part subjected to this potential.
c. Resistance requirements in table V (See 3.18).

4.8.4.1 Preconditioning insulation test samples.- If this test is conducted within 72 hours following the resistance test, they will not have to be preconditioned. However, if the time interval exceeds this period, the preconditioning covered in paragraph 4.8.1.2 will be repeated prior to capacitance.

4.8.4.2 Safety precaution.- Necessary safety precautions shall be taken to protect the test operator from accidentally coming in contact with this high potential.

4.9 Elevation test.- The equipment, placed in its normal operating position in an altitude chamber, shall be tested as follows: (Note: To check temperature stabilization, place a thermocouple on the largest internal mass centrally located within the equipment).

(a) With the equipment placed in the chamber, operating and stabilized at room temperature, take the following measurements:

Paragraph	Measurement
4.8.4, 4.8.4.1 and 4.8.4.2	Insulation test

(b) Reduce the barometric pressure to 20.6 ± 0.1 inches of mercury (simulated 10,000 feet above sea level) and, after temperature and pressure stabilization, repeat the measurements specified in 4.9(a). The equipment shall meet specified performance for those measurements.

(c) Remove the power from the equipment close cover latch, and further reduce the barometric pressure to 3.4 ± 0.1 inches of mercury simulated 50,000 feet above sea level. Maintain this condition for 1 hour.

(d) Return the chamber to ambient pressure. Apply power to the equipment and, after stabilization, repeat the measurements specified in 4.9(a). The equipment shall meet specified performance for these measurements and visual inspection of case and cover.

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4.10 Immersion test.- The equipment, as prepared for field transportation, shall be immersed to a minimum depth of 3 feet of fresh water for 2 hours. Immediately prior to immersion, the temperature of the equipment shall be 40°F, or more, above the temperature of the water. The tank in which the equipment is immersed shall be of sufficient capacity to maintain the water within ±2°F of its initial temperature, or the temperature of the water shall be maintained within those limits by other means. After completion of the 2-hour period of immersion, the equipment shall be removed from the water and wiped dry on exterior surfaces. When the equipment is opened, there shall be no evidence of leakage.

4.11 Temperature test.- The equipment shall be subjected to the temperature cycle shown on Standard MIL-STD-169 except as follows:

a. Remove as soon as possible after completion of Step 8, (-40°F).
b. Place immediately in 0°F (+0, -6°F) test chamber.
c. Retain in chamber for 8 hours (Step 8A)
d. Measurements shall be performed, at 0°F, at the end of at least 8 hours at 0°F.
e. Omit Step 9
f. Conduct Step 10.
g. If feasible, conduct the 0°F step (Step 8A) in lieu of Step 9 in the temperature cycling chamber and delete Steps a through f.

Performance shall be as follows:

Step	Paragraph	Measurement	Performance
4	3.18, and 3.18.1	Resistance	1 ea. random position on each switch to meet requirements of 3.18 and 3.18.1
5	3.18, and 3.18.1	Resistance	Ditto (same switch position)
8A	3.18, and 3.18.1	Resistance	Ditto (same switch position)
10	3.18, and 3.18.1	Resistance	Ditto (same switch position)

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4.12 Moisture-resistance test.- The equipment shall be tested as follows:

4.12.1 Test conditions.-

(a) Do not move equipment from the humidity chamber for measurements.

(b) Start measurements not more than 5 minutes after power is applied to the equipment. Complete measurements as rapidly as possible. Do not leave power on after measurements have been completed.

(c) Perform last two cycles with chassis removed from its enclosure. Cover to be open on first three cycles.

4.12.2 Test method.-

(a) Dry at 130° ±5°F for 24 hours.

(b) Condition at 77° ±5°F and 40 to 50 percent relative humidity for 24 hours.

(c) Subject to continuous cycling for 5 48-hour cycles. Temperature, relative humidity, and period of time for each portion of the cycle shall conform to Standard MIL-STD-170. The measurements specified in 4.12.3 shall be made during the periods shown on the standard and shall comply with 4.12.3.

(d) After cycling has been completed, condition the equipment for 24 hours at 77° ±5°F and 40 to 60 percent relative humidity. Then adjust for optimum performance, using only those means provided by the equipment. No repair or replacement of parts shall be made. After adjustment, the equipment shall meet full specified performance for those measurements specified in 4.12.3.

4.12.3 Performance.- Measurements made during test and specified performance during cycling shall be as follows:

Paragraph	Measurement	Performance during cycling
3.18 and 3.18.1	Resistance	1 ea. Random Position on each switch to meet requirements in 3.18 and 3.18.1
3.19	Adjacent Stator Terminal Insulation	Ditto(Same switch position)
4.29	Visual Inspection	Table IV

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4.12.4 Failure.- If any equipment fails to meet the performance specified in 4.12.3 during cycling, it shall be readjusted once. If the equipment then fails to meet 4.12.3 or fails subsequently during cycling, it does not pass the test. In addition, if the equipment fails to meet full specification requirements after conditioning and adjustment, as specified in 4.12.2(d), it does not pass the test.

4.13 Rain test.- The equipment shall be tested as follows.

4.13.1 Dry at 130° ±5°F for 24 hours.

4.13.2 Condition for 4 hours at 77° ±5°F and 40 to 60 percent relative humidity.

4.13.3 Take the following measurements to insure that the equipment meets specified performance.

Paragraph	Measurement
3.18 and 3.18.1	*Resistance
3.19	*Adjacent Stator Terminal Insulation

*Per paragraph 4.12.3

4.13.4 Place the equipment, in its normal operating position (with cover open), in a test chamber maintained at 77° ±5°F. Allow simulated rainfall to fall on exposed surfaces of the equipment. The water shall be at a temperature of 65° ±5°F and shall fall freely at a rate of 2 inches per hour. Maintain this condition for 1 hour.

4.13.5 With the rainfall continuing at the 2-inch rate, introduce a simulated wind blowing horizontally at 20 ±5 miles per hour against the panel. The water shall fall on exposed surfaces and be blown by the wind against the panel. Maintain this condition for 1 hour.

4.13.6 Immediately after completion of the foregoing, repeat the measurements specified in 4.13.3 without removal from the chamber but with the rain and wind discontinued. Complete the measurements within 1 hour. The equipment shall meet specified performance.

4.13.7 Open the equipment. There shall be no visible evidence of leakage.

4.14 Shock test, bench-handling.- The front-panel assembly shall be removed from its enclosure, as for servicing, and placed in a suitable position for servicing on a solid, 2-inch fir bench top. The test shall be performed as follows, in a manner simulating shocks liable to occur during servicing.

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(a) Tilt up the assembly through an angle of 30 degrees, using one edge of the assembly as an axis, and permit the assembly to drop back freely to the horizontal. Repeat, using other practicable edges of the same horizontal face as axes, for a total of four drops.

(b) Repeat "(a)", with the assembly resting on other faces, until it has been dropped for a total of four times on each face on which the assembly might practically be placed during servicing.

4.15 Shock test drop.- The equipment, with cover closed and latched, shall be dropped a distance of 3 feet on each face, edge and corner (total of 26 drops). The drops shall be made from a "quick release" hook, or pendulum tester as made by the L.A.B. Corporation, Skaneateles, New York, or equal. The floor or barrier receiving the impact shall be of 2-inch fir backed by concrete or a rigid steel frame.

4.16 Bounce test.- The equipment shall be tested on the package tester, as made by the L.A.B. Corporation, Skaneateles, New York, or equal, as follows:

(a) Cover the tester bed with a panel of 1/2-inch plywood, with the grain parallel to the drive chain. Space sixpenny nails, with the heads below the surface, at 6-inch intervals around all four edges and at 3-inch intervals in a 6-inch square in the center.

(b) Place the equipment in its transit case, if any, prepared for field transport, on the bed of the package tester. Limit the lateral motion, by wooden fences, to not more than 3 inches and not less than 1 inch. Additional barriers may be used to prevent tumbling, provided that the fore-and-aft motion of the equipment against the backstop is not restrained.

(c) Operate the package tester, shafts in phase, for a total of 3 hours at 284 ±2rpm. Turn the equipment at the end of each 30 minutes so it will rest on a new face.

4.17 Vibration tests.-

4.17.1 Operational vibration tests.- The decade shall be rigidly mounted on the vibration table with the cover open. During the vibration the test set shall be energized as follows and monitored

a. The connectors and leads shall simulate actual operation on a test bench.

b. All switches set at zero except the 10-ohm decade. This switch shall be set at the maximum resistance position.

c. Maximum voltage shall be applied in accordance with maximum current listed on front panel.

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4.17.1.1 Variable frequency vibration.- Variable frequency vibration shall be applied as follows.

4.17.1.1.1 Test phase 1.- The test set shall be vibrated with the frequency varying between 10 and 33 cycles per second at a total excursion of 0.06±0.006 inch, the frequency varying uniformly from 10 to 33 cycles per second and returning to 10 cycles per second in approximately 1/2 minute.

4.17.1.1.2 Test phase 2.- The test set shall be vibrated with the frequency varying between 10 and 55 cycles per second at a total excursion of 0.03±0.0016 inch, the frequency varying uniformly from 10 to 55 cycles per second and returning to 10 cycles per second in approximately 1 minute.

4.17.1.1.3 Application of the vibration.- Application of the vibration specified in 4.17.1.1 and 4.17.1.1.2 shall be in each of the following directions.

a. Horizontally, parallel to the major horizontal axis of the test set, for a period of 15 minutes for each frequency range.

b. Horizontally, at right angles to the major horizontal axis, for a period of 15 minutes for each frequency range.

c. Vertically, for a period of 15 minutes for each frequency range.

4.17.2 Non-operational vibration tests.- The decade shall be rigidly mounted on the vibration table with cover closed. The test set will not be energized nor monitored.

4.17.2.1 Fixed frequency vibration.- The test set shall then be vibrated for a period of 3 minutes at each of the four most severe resonant frequencies, at the same amplitude as specified in 4.17.1.1, and in the plane observed to give the most severe reaction.

4.17.2.2 Procedure following vibration.- The test set shall be inspected to ascertain that no fixed parts have become loose or damaged. The test set shall be operated to demonstrate that the performance required in the individual test set specification is maintained.

4.18 Visual and mechanical inspection.- Equipment shall be examined for the defects listed in Standard MIL-STD-252.

4.19 Fungi test.- Immediately after the test set has been subjected to the humidity test it shall be removed from its protective cabinet, unless hermetically sealed therein, or when cover

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or cabinet forms an integral part of the mechanism, and the circuit elements exposed and all parts placed in an incubator in which conditions are maintained at 30°C ±5°C (86°F ±9.0°F), 90 to 100 percent relative humidity with some condensation and at atmospheric pressure. The air should be relatively stagnant although a little ventilation should be permitted to occur by convection and leaks through the cabinet. A ventilating fan should not be used. The parts shall be inoculated with a mixture of fungi spores which include the following species: penicillium luteum, aspergillus niger, aspergillus flavus, trichoderma lignorum T-1 U.S.D.A. (see 6.2). Spores shall be mixed with distilled water and approximately 1/2 ounce atomized (using de Vilbiss type atomizer or equal) into the incubator above the parts but not directly on them. This mixture shall be made up fresh for each test inoculation.

NOTE Care should be taken not o inhale this mixture, and to wash the hands after handling.

The inoculated test set shall be incubated for a period of 28 days. At the end of that time the parts shall be visibly examined for evidence of fungus growth. Any visible growth will be cause for rejection.

4.20 Life test.- A life test shall be conducted for a total of 200 hours at room temperature. If necessary, the test may be conducted on an 8 hours per day, 5 day per week basis. Test conditions and steps shall be as follows.

a. Standard test conditions, as specified herein, shall prevail throughout test.

b. No maintenance shall be required during test.

c. Every fifteen minutes of test, the power shall be disconnected at source and the connectors feeding power into the "HI-LO" binding posts be removed momentarily.

d. During this momentary open circuit, all switches shall be switched from 0 position to position 10 and back to 0 position or, if applicable, vice versa.

e. Except as otherwise specified, all switches shall be set at zero except one and this shall be set at 10.

f. Each decade unit shall be fully energized at maximum current for a total of 16 hours.

g. When all decade units have been energized as specified in test step f, the 1-ohm decade resistance unit shall be set at position 10, fully loaded and the higher resistance decade switches set at zero for remainder of test.

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h. In lieu of open circuit switching every fifteen minutes, during the portion of test covered by step g, switching will be performed in fully energized condition every fifteen minutes. Voltage will be decreased as required to maintain power dissipated below maximum values specified herein.

i. Steps g and h shall be performed for at least 24 hours.

4.21 Standard test conditions.- Unless otherwise specified, all room temperature tests shall be conducted at +77 ± 2°F and less than 60 percent humidity.

4.22 Standard performance test.- Unless otherwise specified, the following test steps shall be performed following all tests to ascertain if resistance has been varied appreciably or any defect has occurred.

a. Resistance check of all nomially attainable values.

b. Applicable visual inspection of component.

This does apply to visual and mechanical inspection, resistance and capacitance tests. Resistance check does have to be conducted after the immersion test unless evidence of leakage is noted inside the decade.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.-

5.1.1 Level A.- Resistor, Decade ZM-16( )/U shall be preserved, packaged and tested in accordance with the procedures specified for the designated methods as prescribed in Specification MIL-P-116 and as described herein.

5.1.1.1 Technical manual.- Each technical manual shall be packaged Method IC-6 as follows. Inclose each technical manual within a close-fitting type II class d bag conforming to Specification MIL-B-117. Heat seal the bag to effect closure.

5.1.1.2 Resistor, Decade ZM-16( )/U.- Each resistor, decade shall be packaged Method III as follows: Place the decade and technical manual (see 5.1.1.1) within two thickness of class 1, type III paperboard cushioning conforming to Specification PPP-P-291. Each sheet of paperboard shall be applied separately and secured with class 3 gummed paper tape conforming to Specification UU-T-111. The wrappings shall be applied with the corrugations running in opposite directions. The end results shall be a snug conforming wrap. Place one each wrapped decade

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within a close-fitting style II, type A, class (a) folding paperboard box, conforming to Specification PPP-B-566. Box closure shall be in accordance with the referenced specification or appendix thereto.

5.1.1.3 Intermediate package.- Place eight each, packaged decade within a close-fitting regular slotted style corrugated fiber box, conforming to class 2 WSc Specification PPF-B-636. Box closure shall be in accordance with the box specification or appendix thereto.

5.1.2 Level C.- Resistor, Decade ZM-16( )/U shall be preserved and packaged in accordance with commercial practice and in a manner that will afford protection against corrosion, deterioration and physical damage during direct shipment to the first receiving activity.

5.2 Packing.-

5.2.1 Level A.- Resistor, Decade ZM-16( )/U, packaged as specified in 5.1.1 shall be packed for shipment within class 2, nailed and lock-corner wood boxes, conforming to Specification PPP-B-621, or class 3, wood wirebound boxes, conforming to Specification PPP-B-585, or wood-cleated fiberboard boxes (overseas type) conforming to Specification PPP-B-591, or wood, cleated-plywood boxes (overseas type) conforming to Specification PPP-B-601, or wood cleated, veneer paper overlaid boxes (overseas type) conforming to Specification MIL-B-10377. The weight of any one box with contents shall not exceed 200 pounds. Boxes shall be fabricated to fit the contents snugly. Box closure shall be in accordance with the applicable box specification or appendix thereto.

5.2.1.1 Metal strapping.- Shipping containers shall be strapped in conformance with the requirements of the appendix of the applicable container specification only for direct shipment to ports.

5.2.2 Level B.- Resistor, Decade ZM-16( )/U shall be packed for shipment as specified in 5.2.1 except boxes shall be domestic class or type. No strapping required.

5.2.3 Level C.- Resistor, Decade ZM-16( )/U shall be packed for shipment in a manner conforming to the requirements of Uniform Freight Classification No. 4 and Consolidated Freight Classification, Rule 21 for rail shipment, National Motor Freight Classification No. A-3 and No. 14 for truck shipment, Parcel Post Regulations, and the regulations of other carriers as applicable to the mode of transportation employed at lowest transportation rate.

5.3 Marking.- Interior packages and exterior shipping containers shall be marked in accordance with the applicable provisions of Standard MIL-STD-129.

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6. NOTES

6.1 Intended use.- The decade is normally used as a fixed, accurate, and conveniently selectable resistance in the calibration or accuracy testing of electrical meters. Usually, it is substituted for the multiplier or shunt resistor of the meter being checked or calibrated, or as a resistance standard when checking or calibrating ohmmeters. It is used in conjunction with electrical test sets, such as TS-682( )/GSM-1. It is limited to direct current (dc) circuits, and alternating current (ac) currents, with frequencies up to approximately 60 cycles per second (cps). Where accurate load resistance, ratio arms in bridges, and whenever test or repair work requires a precision resistor, these decades may be successfully applied.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification and any amendment thereto.

(b) Type required.

(c) Level of packaging and level of packing required for shipment. (Level A, level B, or level C).

(d) The specific paragraphs of section 5 which are applicable to the particular procurement.

(e) Marking and shipping of samples.

(f) Place of final inspection.

(g) Technical literature required. (See 3.25)

(h) Submission of the statement of treatment referenced in 3.14 as soon as possible after award of contract. This statement should be submitted to the contracting officer.

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6.3 Definition of level B.- Level B packaging requirements detailed herein are designed to furnish protection against corrosion, deterioration, and physical damage during domestic shipments, covered storage for a period of three years or less, and distribution within the continental limits of the United States.

6.4 Group C inspection.- Approval to ship may be withheld, at the discretion of the Government, pending the decision from the contracting officer on the adequacy of corrective action. (See 4.5.3.2)

6.5 Nomenclature.- The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design, for example ZM16A/U. The contractor should apply for nomenclature in accordance with the applicable clause in the contract. (See 1.1)

6.6 Verification inspection.- Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product, and will normally be identified by the categories listed below:

(a) Type A—The total of that inspection set forth in the Quality Assurance Provisions of this specification or the contract. Included in this category is that amount of inspection referred to as normal and tightened inspection by Military Standard 105.

(b) Type B—That inspection set forth in the Quality Assurance Provisions of this specification or the contract reduced in amount under the reduced inspection provisions of Military Standard 105.

(c) Type C--A reduced inspection procedure resulting in a material reduction in the amount of inspection set forth in the Quality Assurance Provisions of this specification. The amount of inspection is less than that provided for in type B and is based upon a consistently acceptable product resulting from a planned quality control system voluntarily employed by the contractor in the production of the product.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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